

Transition Amplitudes and Probabilities from Relaxed Cobordisms

Theory, Provenance, and Two Extensions

Tessera — cobordism subsystem

Abstract

We collect, derive from first principles, and critically assess the machinery by which the Tessera cobordism subsystem reads a *transition amplitude* $Z(W) = \langle \psi_A | U | \psi_B \rangle$ off a simplicial cobordism W after a relaxation step. The construction rests on four layers, each of a different epistemic status: (i) discrete Hodge theory, which makes harmonic k -forms the boundary state space (theorem); (ii) the topology of the carried register, which decides *which* boundary states a given bulk can transport (theorem); (iii) the Choi–Jamiolkowski map–state duality, which identifies the operation reading with the pairing (algebraic identity); and (iv) a TQFT gluing law, under which stacked cobordisms compose and amplitudes interfere. We derive the exact transport-amplitude identity $Z_T(q, \alpha) = (T^{-1}q)^\dagger G \alpha$ and its deviation law, locate the emergent gate as $u' = (T^\dagger)^{-1}$, and prove that probability normalization $\sum_b |Z|^2 = 1$ is *exactly* the unitarity of u' . We then specify two extensions that are not yet certified: a normalized-probability experiment (Born rule + coherent composition) and the coupling of the full complex Regge action relaxation to the amplitude readout. Throughout we separate what is a theorem from what is a numerical certificate from what is conjectural, and we flag the conformal runaway and the discretization floor as the load-bearing physical caveats.

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1 Scope and provenance

This document is theory, not a results report. It formalizes the content of two committed capstone examples and the algebraic oracle they lean on:

- `examples/cobordism/dw_spectral_bridge.py` (the v0.4 bridge, #174): three independent readings $Z_{\text{DW}} = \langle \psi_A | U \psi_B \rangle = Z_{\text{spec}}$ on $\Sigma = T^2$;
- `examples/cobordism/level1_value_reading.py` (#280/#284): the anchored transport value $Z_T(q, \alpha)$ and its exact deviation law;
- `include/quantum/ChoiJamolkowski.h`: the dense map–state duality;
- `examples/cobordism/stationary_action_relaxation.py` (#340/#343): the $\Phi = \|\nabla S\|^2 + \Gamma r_U$ objective.

The amplitude bridge uses *spectral* realizability (the Hodge picture), which is metric-light; the full *stationary-action* relaxation is a separate object, and the fact that the two are not yet wired together is the central open problem (§8–§9).

2 The state space: discrete Hodge theory

Let Σ be a finite simplicial complex with a metric (edge squared-lengths ℓ). On k -cochains $C^k = C^k(\Sigma; \mathbb{C})$ the metric induces the weighted inner product through the circumcentric dual volumes (the Hodge weights w_k),

$$\langle a, b \rangle_k = \sum_{\sigma \in \Sigma_k} w_k(\sigma) \overline{a(\sigma)} b(\sigma). \quad (1)$$

The coboundary $d_k : C^k \rightarrow C^{k+1}$, $(d_k a)(\tau) = \sum_{\sigma \subset \tau} [\sigma : \tau] a(\sigma)$, has the Hodge adjoint $\delta_k = W_k^{-1} d_k^\top W_{k+1}$ with respect to (1), where $W_k = \text{diag}(w_k)$. The Hodge Laplacian is

$$L_k = \delta_k d_k + d_{k-1} \delta_{k-1}, \quad (2)$$

which the code assembles in symmetric form (`numpy_L1`):

$$L_1^{\text{sym}} = B_1^\top B_1 + B_2 B_2^\top, \quad B_1 = d_1 W_1^{-1/2}, \quad B_2 = W_1^{1/2} d_2 W_2^{-1/2}. \quad (3)$$

Theorem 2.1 (Discrete Hodge decomposition). *There is an orthogonal decomposition $C^k = \text{im } d_{k-1} \oplus \ker L_k \oplus \text{im } \delta_k$, and the harmonic space $\mathcal{H}^k := \ker L_k$ satisfies $\mathcal{H}^k \cong H^k(\Sigma; \mathbb{C})$, so $\dim \mathcal{H}^k = b_k$, the k -th Betti number.*

Definition 2.2 (Spectral qubit). The boundary state space is $\mathcal{H}_\Sigma := \mathcal{H}^k(\Sigma)$ with the inner product (1) restricted to it. Harmonic forms *are* states; the Hodge inner product *is* the quantum inner product.

For $\Sigma = T^2$, $k = 1$: $b_1 = 2$, so $\mathcal{H}_{T^2} = \mathbb{C}^2$, spanned by the longitude and meridian classes.

Remark 2.3 (Two distinct state spaces). The Dijkgraaf–Witten space $Z_{\text{DW}}(\Sigma) = \mathbb{C}[H^1(\Sigma; \mathbb{Z}_2)] = \mathbb{C}^4$ is the *group-algebra* state space, of dimension $2^{b_1} = 4$, not $b_1 = 2$. It is a strictly larger object than the spectral qubit; §6 relates the two via the “quantized shadow.” They must not be conflated.

3 Cobordism transport: the carried register

Let W be a cobordism with $\partial W = M_0 \sqcup \overline{M_1}$. The inclusions $\iota_a : M_a \hookrightarrow W$ pull harmonic forms back to the boundary, and the period map over a chosen basis of hole-cycles $\{c_i\}$ extracts the register coordinates:

$$\pi_a : \mathcal{H}^k(W) \rightarrow \mathbb{C}^b, \quad (\pi_a h)_i = \oint_{c_i \subset M_a} h = \sum_{e \in c_i} \pm h(e). \quad (4)$$

(For a triangle hole (a, b, c) this is $h(a, b) + h(b, c) - h(a, c)$, the code's `_period`.)

Definition 3.1 (Carried class). A boundary class is *carried* by W iff it lies in the image of the restriction $H_k(\partial W) \rightarrow H_k(W)$.

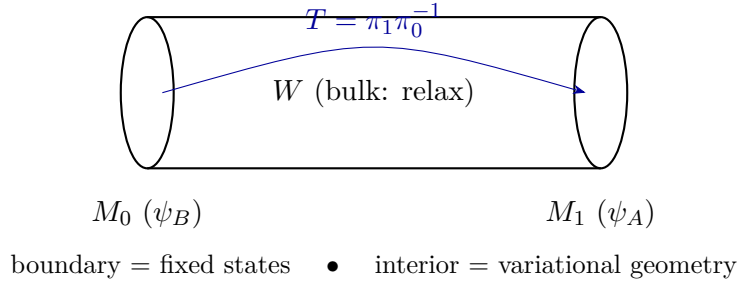
Theorem 3.2 (Longitude/meridian dichotomy). *For the solid torus $W = D^2 \times S^1$ one has $H_1(W) = \mathbb{Z}\langle \text{longitude} \rangle$, while the meridian bounds the disk D^2 and so maps to 0 in $H_1(W)$. Hence the longitude is carried (spectrally realizable, residual $\rightarrow 0$) and the meridian is obstructed (its realizability residual floors).*

This is pure algebraic topology: the realizability oracle's verdict is computing the rank of the restriction map, not a numerical accident. The continuous family $\cos(s) \text{ long} + \sin(s) \text{ mer}$ interpolates between the two and the floor lifts monotonically (the `spectral_boundary_sweep`).

Definition 3.3 (Transport). For a *graph-like* fill (both ends carry the full register, so π_0, π_1 are isomorphisms) the period-level transport is the composite

$$T := \pi_1 \circ \pi_0^{-1} : \mathbb{C}^b \rightarrow \mathbb{C}^b, \quad (5)$$

sending end-0 periods to end-1 periods.



4 The chart, the Gram, and the exact amplitude identity

Use end-0 periods as a coordinate chart on $\mathcal{H}^k(W)$. Define the two named harmonics

$$g_{\text{in}}(\alpha) := \pi_0^{-1}(\alpha), \quad g_{\text{out}}(q) := \pi_1^{-1}(q) = \pi_0^{-1}(T^{-1}q) = g_{\text{in}}(T^{-1}q). \quad (6)$$

The Hodge inner product, expressed in the end-0 chart, is a Hermitian Gram form $\tilde{G}$, $\langle \pi_0^{-1}(x), \pi_0^{-1}(y) \rangle = (1/s_1) x^\dagger \tilde{G} y$, where $G = s_1 \tilde{G}$ is the *anchor-normalized* Gram and the anchor is fixed by one reference state, $s_1^{-1} = \langle g_{\text{in}}(\beta_0), g_{\text{in}}(\beta_0) \rangle$. The transport value is the anchored cross-end pairing,

$$Z_T(q, \alpha) := s_1 \langle g_{\text{out}}(q), g_{\text{in}}(\alpha) \rangle. \quad (7)$$

Proposition 4.1 (Exact transport amplitude). *With the conventions above,*

$$\boxed{Z_T(q, \alpha) = (T^{-1}q)^\dagger G \alpha} \quad (8)$$

Proof. Substitute $g_{\text{out}}(q) = g_{\text{in}}(T^{-1}q)$ from (6) into (7) and use the Gram identity: $Z_T = s_1 \langle g_{\text{in}}(T^{-1}q), g_{\text{in}}(\alpha) \rangle = s_1 \cdot (1/s_1)(T^{-1}q)^\dagger G \alpha = (T^{-1}q)^\dagger G \alpha$. $\square$

Definition 4.2 (Hand amplitude and the emergent gate). Let the *hand amplitude* be $\text{amp}(q, \alpha) := (T^{-1}q)^\dagger \alpha = q^\dagger (T^\dagger)^{-1} \alpha = \langle q | u' \alpha \rangle$, which defines the emergent gate $u' = (T^\dagger)^{-1}$.

Corollary 4.3 (The deviation law). *Subtracting,*

$$Z_T(q, \alpha) - \langle q | u' \alpha \rangle = (T^{-1}q)^\dagger (G - I) \alpha = \tilde{w}^\dagger (G - I) \tilde{\alpha}, \quad (9)$$

with $\tilde{w} = T^{-1}q$ the end-0 period coordinates of $g_{\text{out}}(q)$ and $\tilde{\alpha}$ those of the input. This is exactly the law verified in `level1_value_reading.py`.

Corollary 4.4 (H3: isometric charts). *If the period chart is Hodge-orthonormal, $G = I$, then $Z_T(q, \alpha) = \langle q | u' \alpha \rangle$ identically, for any T . The equivariant (C_3 -symmetric, no diagonal choices) prism achieves $G = I$ at machine precision because the end symmetry extends into the bulk; the staircase prism's order-dependent diagonals break it, giving $\|G - I\| \sim 10^{-2}$.*

Remark 4.5 (Two independent conditions). Isometry of the chart ($G = I$, so Z_T reads cleanly) and unitarity of the transport ($T^\dagger T = I$, equivalently $u' = (T^\dagger)^{-1} = T$) are *logically independent* properties of a fill. A clean unitary quantum amplitude needs both. This separation drives the two gates of Experiment A (§9).

5 The operation reading: Choi–Jamiołkowski

The Choi–Jamiołkowski isomorphism bends an operator into a state, $\text{vec}(U) = \sum_{ij} U_{ij} |i\rangle \otimes |j\rangle$, with $\text{vec}(|a\rangle\langle b|) = a \otimes \bar{b}$ on a rank-one outer product. The central map–state (Hilbert–Schmidt) duality is an algebraic identity:

$$\langle \psi_A | U | \psi_B \rangle = \langle \text{vec}(U_T), \text{vec}(U) \rangle = \text{Tr}(U_T^\dagger U) = \sum_{ij} \overline{\psi_{A,i}} U_{ij} \psi_{B,j}, \quad U_T = |\psi_A\rangle\langle\psi_B|. \quad (10)$$

Because (10) holds with no reference to geometry, the “operation” reading is a trustworthy oracle against which the spectral reading is checked. In `dw_spectral_bridge.py` the three readings

$$Z_{\text{DW}}(W) = \langle \psi_A | U | \psi_B \rangle = Z_{\text{spec}} \quad (11)$$

agree to 10^{-7} on longitude-aligned states, each cross-checked against an independent `numpy` reimplementation (a GF(2) state sum, a direct `vdot`, and a $\ker L_1$ projection).

6 Gluing and coherent composition

The framework is a discrete Atiyah TQFT: a closed $(d-1)$ -manifold Σ gets a space $\mathcal{H}_\Sigma$, a cobordism $W : \Sigma_0 \rightarrow \Sigma_1$ gets a linear map $Z(W)$, and

$$W = W_2 \circ_{\Sigma_1} W_1 \Rightarrow Z(W) = Z(W_2) Z(W_1), \quad \mathcal{H}_{\Sigma \sqcup \Sigma'} = \mathcal{H}_\Sigma \otimes \mathcal{H}_{\Sigma'}, \quad Z(\emptyset) = \mathbb{C}. \quad (12)$$

Stacked fills compose (verified to 10^{-9}). Consequently amplitudes compose *coherently* — the sum over intermediate states happens *before* the modulus:

$$Z(b, a) = \langle b | Z(W_2) Z(W_1) | a \rangle = \sum_m Z_2(b, m) Z_1(m, a), \quad (13)$$

$$P(b|a) = \left| \sum_m Z_2(b, m) Z_1(m, a) \right|^2 \neq \sum_m |Z_2(b, m)|^2 |Z_1(m, a)|^2. \quad (14)$$

The difference between (14) and the classical (Markov) right-hand side is the interference cross-term $2 \operatorname{Re} \sum_{m \neq m'} Z_2(b, m) Z_1(m, a) \overline{Z_2(b, m')} Z_1(m', a)$: the operational signature of a genuine quantum amplitude.

Remark 6.1 (The quantized shadow). The Dijkgraaf–Witten invariant is the state sum $Z_{\text{DW}}(W) = \sum_{\phi \in \text{flat } \mathbb{Z}_2\text{-conn.}} \prod_{\text{simplices}} \omega(\phi)$. For trivial or sign cocycle the values are integer-quantized, so $\operatorname{im}(Z_{\text{DW}}) \subset \operatorname{Hom}(\mathcal{H})$ is a *discrete lattice*. The spectral Z is a continuum (it realizes the longitude and a neighborhood). Hence $Z_{\text{DW}} = Z_{\text{spec}}$ only at the lattice points; a generic U leaves it continuously, the gap opening as U departs the discrete DW image. The topological theory is the integer skeleton of the spectral one.

7 Born normalization is exactly unitarity

Proposition 7.1 (Normalization $\Leftrightarrow$ unitarity). *Let $\{b_k\}$ be an orthonormal basis of the output state space, so $\sum_k |b_k\rangle\langle b_k| = I$. On an isometric fill ($G = I$, so $Z_T(b_k, \alpha) = \langle b_k | u' \alpha \rangle$),*

$$\sum_k P(k|\alpha) = \sum_k |\langle b_k | u' \alpha \rangle|^2 = (u' \alpha)^\dagger \left(\sum_k |b_k\rangle\langle b_k| \right) (u' \alpha) = \langle \alpha | u'^\dagger u' | \alpha \rangle. \quad (15)$$

Therefore $\sum_k P(k|\alpha) = \|\alpha\|^2$ for all α iff $u'^\dagger u' = I$, i.e. iff the emergent gate is unitary. Equivalently, since $u' = (T^\dagger)^{-1}$, iff $T^\dagger T = I$.

So probability conservation is not an extra postulate to be imposed — it is a *theorem conditioned on a unitarity certificate*. The two hypotheses $\{G = I, T^\dagger T = I\}$ are the complete set; verify them and the Born rule follows. When $G \neq I$ the correct sum is $\langle \alpha | u'^\dagger G u' | \alpha \rangle$, and (9) accounts for the deficit exactly: a non-conserving “probability” is then a *measured chart artifact*, not a hidden leak.

8 The action and backreaction

Geometry enters through the weights. The Lorentzian (Sorkin) Regge action is

$$S(\ell) = \sum_{\text{hinges } h} A_h(\ell) \varepsilon_h(\ell) \in \mathbb{C}, \quad (16)$$

with A_h the codimension-2 hinge volume and ε_h the deficit angle, which is *complex* in Lorentzian signature (spacelike hinges contribute boosts via acosh , hence imaginary parts). Since the Hodge weights $w_k(\ell)$ are dual volumes, every downstream object is a functional of ℓ :

$$\ell \longrightarrow L_k(\ell) \longrightarrow \mathcal{H}^k(\ell) \longrightarrow T(\ell), G(\ell) \longrightarrow u'(\ell), Z_T(\ell). \quad (17)$$

Proposition 8.1 (How relaxation moves the amplitude). *For $L_k(\ell)h = 0$ a metric perturbation $\delta\ell$ moves the harmonic by degenerate first-order perturbation theory,*

$$\delta h = -L_k^+ (\delta L_k) h, \quad \delta Z_T = s_1 (\langle \delta g_{\text{out}}, g_{\text{in}} \rangle + \langle g_{\text{out}}, \delta g_{\text{in}} \rangle) + (\delta s_1) \langle g_{\text{out}}, g_{\text{in}} \rangle, \quad (18)$$

where L_k^+ is the Moore–Penrose pseudoinverse on $\operatorname{im} L_k$. Thus $dZ_T/d\Gamma$ is computable and is the backreaction signal.

The relaxation objective is

$$\Phi(\ell) = \|\nabla S(\ell)\|^2 + \Gamma r_U(\ell), \quad (19)$$

each term load-bearing:

- $\|\nabla S\|^2$ drives $\delta S = 0$ — one *extremizes*, never minimizes. The bare action is unbounded below (the conformal mode $g \mapsto e^{2\varphi}g$ has a wrong-sign kinetic term, hence a runaway), so the stationary point is the target. One keeps $\text{im } S$: minimize $\sum |\partial S|^2$, not $\sum (\partial \text{Re } S)^2$, because the imaginary part fixes the overall scale the real part leaves loose.
- $r_U = \text{residualForPeriods}$ is the *register fidelity*, $r_U = \|\text{periods}(\text{relaxed harmonic}) - \text{target}\|$. It is the Lagrange-type penalty enforcing $\partial W = (\text{prepared states})$ *under* relaxation: the geometry may evolve only while the boundary state stays carried. This is the cobordism evolution rule made into an objective.

Caveat. Do not sidestep the conformal runaway. The mass that regulates the runaway to a convergent interior minimum *is* the backreaction physics: no linearizing, boundary-pinning, or freezing of the matter field, and no swapping the energy functional.

Remark 8.2 (The deep open question). Does $u'(\ell^*)$ stay unitary after action relaxation? Relaxation generically breaks the equivariance that gave $G = I$, so $G(\ell^*) \neq I$ and possibly $u'(\ell^*)^\dagger u'(\ell^*) \neq I$. By Proposition 7.1 that would be *backreaction-induced apparent non-unitarity* — probability flowing from the register into the geometry. Whether the stationary condition $\delta S = 0$ *protects* unitarity (a discrete analogue of unitarity from a real action / reflection positivity) or breaks it is exactly what Experiment B would measure. Either answer is a result.

9 Two extensions to certify

Experiment A — normalized transition probabilities. Substrate: an *isometric* graph-like transport fill (CellsFill on the equivariant C_3 prism), where both end blocks P_0, P_1 are full rank so the whole register is carried.

1. Read $u' = \text{emergent_gate}()$; confirm $\|G - I\| < 10^{-9}$ (chart isometric).
2. **Unitarity gate** (the normalization certificate): $\|u'^\dagger u' - I\| < 10^{-9}$.
3. For random unit inputs α , verify $\sum_k |Z_T(b_k, \alpha)|^2 = 1$ to 10^{-9} (Proposition 7.1).
4. **Born = hand:** $|Z_T(b_k, \alpha)|^2 = |\langle b_k | u' \alpha \rangle|^2$.
5. **Coherent vs. classical:** confirm P obeys (14), not the Markov sum — the interference cross-term is nonzero (the quantum discriminator).
6. **Negative control:** on an anisotropic fill, $\sum_k P \neq 1$ and the deficit equals the $(G - I)$ correction of (9) — leakage named, not hidden.

Experiment B — action-coupled amplitude readout. Substrate: a transport fill (not a merge — merge operator readout is deferred, #376/#380), boundary edges pinned to encode ψ_A, ψ_B , interior squared lengths variational.

1. Relax $\Phi(\ell) = \|\nabla S\|^2 + \Gamma r_U$ (reuse StationaryActionRelaxation; keep $\text{im } S$).
2. At ℓ^* , rebuild $\mathcal{H}^1(\ell^*)$, recompute $g_{\text{in}}, g_{\text{out}}, s_1$ and read $Z_T(q, \alpha; \ell^*)$ via Proposition 8.1.
3. **Reduction:** $\Gamma \rightarrow 0$, uniform start $\Rightarrow Z_T(\ell^*) = Z_T(\text{uniform})$.
4. **Stationarity:** $\|\nabla S\|^2(\ell^*)$ at its intrinsic floor; **register survived:** $r_U(\ell^*) < \text{tol}$.
5. **Backreaction signal:** sweep Γ ; record the continuous shift of $Z_T(\ell^*)$ off the flat value.
6. **Unitarity under backreaction:** test $u'(\ell^*)^\dagger u'(\ell^*) = I$, with any breakage $(G - I)$ -accounted.

Sequencing: A before B. A certifies the probability machinery on a clean substrate; B then asks whether unitarity survives action relaxation. Doing B first means debugging the relaxation and the Born normalization simultaneously.

10 Epistemic ledger

Statement	Status
$\mathcal{H}^k(\Sigma) \cong H^k(\Sigma; \mathbb{C})$; states are harmonic forms (Thm. 2.1)	theorem
Longitude carried / meridian obstructed (Thm. 3.2)	theorem (topology)
$Z_T = (T^{-1}q)^\dagger G\alpha$ and the deviation law (Prop. 4.1, Cor. 4.3)	theorem (derived here)
H3: $Z_T = \langle q u'\alpha \rangle$ when $G = I$ (Cor. 4.4)	theorem; verified 10^{-9}
Choi map–state identity (10)	algebraic identity
TQFT gluing / coherent composition (12)–(14)	theorem; composition verified 10^{-9}
$Z_{\text{DW}} = \langle \psi_A U \psi_B \rangle = Z_{\text{spec}}$ on the DW-representable set	verified 10^{-7} ; <i>discrete</i> set only
$\sum_k P = 1 \Leftrightarrow u'$ unitary (Prop. 7.1)	theorem, <i>conditional</i> on the certificate
u' is unitary on the equivariant fill (Exp. A, step 2)	not yet certified
Coherent (non-Markov) composition of P (Exp. A, step 5)	not yet certified
Z_T read from the action-relaxed ℓ^* (Exp. B)	bridge does not yet exist
$\delta S = 0$ protects unitarity under backreaction	conjectural

The solid foundation is §2–§6 (theorems and algebraic identities). Proposition 7.1 is a theorem *conditional on* a certificate that has not been run. The backreaction coupling of §8 is mechanically sound but physically untested. The two load-bearing physical caveats are the conformal runaway (the regulator is the physics, not a convenience) and the intrinsic discretization floor ($\sim 4 \times 10^{-2}$ intertwine residual on non-equivariant metrics), which a symmetry-invariant metric does not remove.